

MedGuard AI – Project Development Plan

● Phase 1 – Requirement Analysis & Planning****Objectives:

- Finalize system scope
- Freeze technology stack (Django, PostgreSQL, FAISS, Ollama, Docker)
- Define architecture and modules

Activities:

- Study problem statement
- Identify functional & non-functional requirements
- Define system constraints
- Risk analysis planning
- Dataset collection planning (medicine + legality data)

Deliverables:

- ☒ Final SRS Document
- ☒ Architecture Blueprint
- ☒ Risk Assessment Plan

● Phase 2 – System Architecture & Database Design****Objectives:

- Design microservice architecture
- Design PostgreSQL schema in 3NF

Activities:

- Define microservices:
 - API Gateway
 - Medicine Verification Service
 - Symptom LLM Service
 - PostgreSQL DB
 - FAISS persistent volume
- Design tables:
 - Users
 - Medicines
 - Medicine Barcodes
 - Embeddings Metadata
 - Verification History
 - Symptom History

- Define REST communication flow

Deliverables:

- ☒ ER Diagram
- ☒ Logical Database Design
- ☒ Service Communication Diagram

● Phase 3 – Dataset Preparation & RAG Setup****Objectives:

- Prepare medicine dataset
- Implement RAG pipeline

Activities:

- Clean and normalize medicine dataset
- Assign legality ENUM values
- Generate embeddings using Sentence Transformers

- Build FAISS index
- Store embedding metadata in PostgreSQL
- Configure persistent FAISS volume

Deliverables:

- ☒ Cleaned Medicine Dataset
- ☒ FAISS Vector Index
- ☒ Embedding Metadata Records

**** Phase 4 – Backend Development (Core Microservices)** ♦ 4.1 API Gateway Development**

- JWT authentication
 - User registration & login
 - Request routing
 - Error handling
 - ♦ 4.2 Medicine Verification Service
 - OCR integration
 - Barcode decoding
 - RAG similarity search
 - Expiry verification logic
 - Legality classification
 - Authenticity decision logic
 - LLM-based summary generation
 - ♦ 4.3 Symptom Remedy LLM Service
 - Ollama integration
 - Strict prompt engineering
 - Safe output formatting
 - Disclaimer enforcement
- Deliverables:**
- ☒ Functional REST APIs
 - ☒ Inter-service communication working
 - ☒ Authentication system

Phase 5 – Frontend Development**Objectives:**

- Develop UI (React / Flutter)
- Features:**
- Multi-modal medicine input (Text/Image/Barcode)
 - Result classification display
 - Color-coded status (Authentic / Fake / Expired / Illegal)
 - Symptom input interface
 - Disclaimer display
 - History dashboard
- Deliverables:**
- ☒ Responsive Web UI
 - ☒ Android App (Initial Phase)

Phase 6 – Dockerization & Deployment**Objectives:**

- Containerize all services
- Enable scalable deployment

Activities:

- Create Dockerfiles
- Configure Docker Compose
- Set up:
 - api-gateway
 - medicine-service
 - symptom-llm-service
 - postgres-db
 - ollama
- Configure persistent FAISS storage
- Deploy to AWS / DigitalOcean

Deliverables:

- ☒ Dockerized Microservice System
- ☒ Cloud Deployment Setup
- ☒ Persistent Volumes Configured

● Phase 7 – Testing & Validation**** ♦ Functional Testing

- Multi-modal input validation
- Expiry detection accuracy
- Legality classification
- RAG similarity correctness
 - ♦ **Performance Testing**
- FAISS search under 5 seconds
- 100 concurrent users test
- LLM latency measurement
 - ♦ **Security Testing**
- JWT validation
- SQL injection prevention
- File upload validation
 - ♦ **Risk Testing**
- LLM hallucination checks
- Similarity threshold tuning
- Fault isolation testing

Deliverables:

- ☒ Test Cases Document
- ☒ Performance Report
- ☒ Bug Fix Logs

● Phase 8 – Optimization & Hardening****Activities:

- Tune FAISS index
- Optimize embedding dimension
- Improve Docker scaling
- Implement caching (if required)
- Optimize database indexing
- Add logging & monitoring

Deliverables:

- ☒ Optimized Search Performance
- ☒ Stable Production Version

Phase 9 – Documentation & Final Deployment**Activities:**

- API documentation
- Deployment guide
- User manual
- Admin manual
- Prepare viva explanation (RAG, FAISS, microservices)

Deliverables:

-  Final Project Report
-  Deployment Manual
-  Demo-Ready System

Phase 10 – Future Enhancements (Post-Release)

As defined in SRS:

- Government API integration
- Pharmacy dashboard
- Counterfeit reporting
- Multi-language support
- Confidence score visualization
- Advanced AI health assistant
- iOS support